obtained by pressing the touch panel using a stylus (or finger). If the pressing point is closer to the left contact of X surface, the voltage will be closer to 0V and the controller will determine the exact position of the point depending on the voltage obtained.

Similarly, in order to determine Y coordinate, it is necessary to connect the bottom contact on Y surface to ground and the upper contact to +5V power supply.

The 128x64 LCD is divided into two equal halves, with each half being controlled by a separate KS0108 controller. Such LCDs (using KS0108 controllers) involve a paging scheme, that is, the whole LCD is divided equally into pages. Paging scheme of the graphical LCD can be understood from Table I.

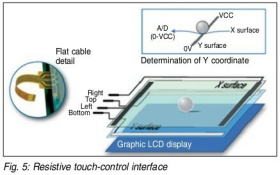


Fig. 5: Resistive touch-control interface

Keyboard interface module. The keyboard interface module circuit is shown in Fig. 6. This module includes an ATmega8 microcontroller (IC8), keyboard input, 7805 regulator (IC7), 16x2 LCD (JHD162A) and a few other components. The input given from the keyboard is displayed on the LCD, and is simultaneously transmitted to the main module through connector CON9.

This module is basically for doing various settings like date and time, testing of each module and turning off the system using a specific command. It thus helps to set the system to correct values and operating parameters. This module is not used during normal operation and may therefore be kept disconnected.

A 16x2 LCD can display 16 characters per line and there are two such lines.

TABLE I
Paging Scheme of the GLCD

	Controller1 CS=1, CS=0	Controller2 CS=0, CS=1
D0 D1 D2 D3 D4 D5 D6 D7	Page0 64 bytes (columns)x8bits	Page 0
	Page 1	Page 1
	Page 2	Page 2
	Page 3	Page 3
	Page 4	Page 4
	Page 5	Page 5
	Page 6	Page 6
	Page 7	Page 7
	64 bits	64 bits

64 bits

Software program

The program for the microcontroller is written in C language and compiled using gcc compiler. The compiler is used to convert C program into a hex file for burning into the flash memory of ATmega8 microcontroller.

In this project, the default internal oscillator is used for ATmega8 operation. The complete program is a combination of different functions. There are four main programs for this

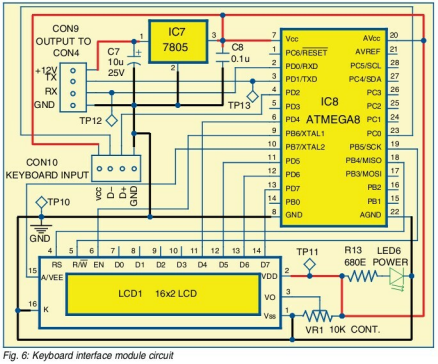


Fig. 6: Keyboard interface module circuit

TABLE II
Test Points

Test point	Details
TP0, TP6, TP14, TP10	0V (GND)
TP1, TP3, TP7, TP11, TP15	+5V
TP2	+3.3V
TP4, TP9, TP13, TP16	Train of pulses when ATmega8 transmits data
TP5, TP8, TP17, TP12	Train of pulses when ATmega8 receives data